



The impact of product management on SME firm performance

SME firm
performance

85

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Abstract

Purpose – This study seeks to address a gap in the literature by investigating product management (PM) as a set of firm-level activities, distinct from the behaviours embedded within the market orientation (MO) construct. This research establishes PM as a set of organizational activities, which lie at the boundary between the traditional functions of the firm.

Design/methodology/approach – A model is proposed and tested using a heterogeneous sample of 316 Canadian small and medium-sized enterprises, where PM mediates the relationship between MO and firm performance. Data were analyzed using a partial least squares, structural equation model.

Findings – PM behaviour is found to fully mediate the MO – firm performance relationship. Two of the three PM constructs, namely channel analysis/support and market/technical integration account for this effect.

Research limitations/implications – These results support the PM literature, which infers that more emphasis on external boundary spanning activities and internal coordination should positively influence firm performance. Limitations include the heterogeneous nature of the sample, time frame and geographic bias.

Practical implications – Managerial implications include the establishment of an empirical link between PM practices and how firms ultimately perform. This could assist practitioners in enhancing coordination activities between the marketing and technological factions within the organization.

Originality/value – This research establishes PM as a set of activities, which lie at the boundary between the traditional functions of the firm. These activities are found to fully mediate the MO – firm performance relationship and introduce the hereto untested link between PM and firm performance.

Keywords Market orientation, Small to medium-sized enterprises, Marketing/entrepreneurship interface symposia, Product management, Boundary spanning, Partial least squares, Firm performance, Canada

Paper type Research paper

Introduction

Product management (PM) as a concept has a long association with the marketing function of the firm, dating back to its auspicious beginnings in the 1930s when Procter & Gamble (P&G) successfully deployed this management model throughout their organization (Eckles and Novotny, 1984; Lysonski, 1985; Wood and Tandon, 1994; Wichman, 1986). This system, which treated the product as the focal point of the management structure, delegated the entire responsibility for the development, production and marketing of a product to an individual, known as a product manager. Under this system, the firm concentrated the responsibility for the integration of all functions required for the successful creation though to marketing of a product or product line. Although this system was never widely adopted by small and medium-sized enterprises (SMEs), successful SMEs appear to display both formal and informal PM strategies (Karlsson and Olsson, 1998), likely due to their simpler



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organizational structures (Becherer *et al.*, 2003) and general flexibility (Pelham, 1999; Pelham, 2000). These behaviours have been cited as some of the most crucial activities for product success in SMEs (Karlsson and Olsson, 1998).

Why then has PM research mostly evaded the focus of academic research? This may be due to this business philosophy pre-dating most academic management research, which came into favour in the 1960s just as the classic PM model was being replaced with more sophisticated views of the firm. This modern era of management favoured the specialization of functions rather than the integration of functional of activities. Thus, the boundary spanning activities traditionally associated with the PM role became generally relegated to the informal systems within the organization. Although many of the classic PM behaviours are still embedded within organizations, PM as a research area has mostly languished. With notable exceptions (Tyagi and Sawhney, 2010; Katsanis, 2006), most recent publications can best be categorized as practitioner-based (e.g. Gorchels, 2005).

Interestingly, market orientation (MO), long believed to be a driver and predictor of firm performance, remained largely subjective in nature until the influential work of Narver and Slater (1990), Jaworski and Kohli (1993) and Deshpandé *et al.* (1993) in the early 1990s. This research was successful at quantifying the marketing-related culture and behaviour of the firm and established the basis for a deep and rich body of academic research which continues to develop. Although several researchers have since investigated MO's affect on performance within SMEs (Gray *et al.*, 1998; Pelham, 1997; Verhees and Meulenbergh, 2004), its impact is somewhat inconclusive and remains largely context dependant (Greenley, 1995; Pelham and Wilson, 1996; Becherer *et al.*, 2001; Frishammar, 2009). However, none have investigated the boundary spanning behaviour traditionally associated with the PM activities of the firm.

Although PM has lost much of its lustre (Wood and Tandon, 1994), it continues to have its roots entrenched within the marketing function of the firm. This relationship however has not been investigated in any empirical way, likely due to the implicit and largely false assumption that these behaviours are embedded within the well-established MO construct. However, unlike MO, PM establishes how the firm's behaviours transcend the traditional functional boundaries of the firm and how external information is captured and processed by the organization. MO's more simplistic relationship does not take into account the complex set of internal and external interactions that a firm must manage in order to optimize performance. This research thus addresses a gap in the literature by taking the position that these firm-level PM activities lie at the boundary between the traditional functions of the firm. PM can thus be examined as a set of firm behaviours, which enhance or mediate the MO – firm performance relationship. This research thus addresses some of the issues at the marketing/entrepreneurship interface (Miles and Darroch, 2008; Ulsay and Teach, 2008; Hansen and Eggers, 2010), namely product positioning, channel management and cross functional coordination within SMEs. This study thus seeks to answer what types of PM behaviours SMEs display and how they affect performance.

The literature on PM and MO is reviewed in the context of SMEs. A model is then proposed and hypotheses developed linking PM with MO and firm performance. The model is then tested using a heterogeneous sample of 316 Canadian small and medium-sized firms in the manufacturing sector. Results are discussed in the context of both academic and managerial implications.

Literature review

Product management literature

The extant PM literature largely focuses on the functions of the individual or set of individuals who are responsible for stewardship of the product within the organization. These individuals, known as product managers, span the traditional boundaries of their firm and act as integrators and disseminators of information used for product-related decision making. Not to be confused with product development or project management, classic PM has a long association with the marketing function of the firm, dating back to the 1930s when P&G first implemented the role within its divisions as a separate business function (Eckles and Novotny, 1984; Lysonski, 1985; Wood and Tandon, 1994; Wichman, 1986). However, unlike the firm's marketing function, PM involves the integration of all the functions required for the successful creation through to market adoption of a product, including complete accountability for its creation, production and marketing. This integrated "conception through to purchase" system became widely adopted by industry until the 1970s when multifunctional teams began to replace individual product managers (Katsanis *et al.*, 1996). PM within the enterprise thus became relegated to sub-ordinate specialty positions, which came to include segment manager, product line manager or product business manager (Wichman, 1986; Tyagi and Sawhney, 2010).

Topics within the PM literature range from differences between industrial and consumer sectors (Eckles and Novotny, 1984; Quelch *et al.*, 1992; Murphy and Gorchels, 1996; Gemmill, 1972; Lysonski and Woodside, 1989; Dawes and Patterson, 1988; Cummings *et al.*, 1984; Kelly and Hise, 1979), the nature of the responsibilities of the function (Dominguez, 1971; Lehmann and Winer, 2005; Clewett and Stasch, 1975) and the agent-firm relationship (Lysonski, 1985; Murphy and Gorchels, 1996; Cossé and Swan, 1983; Giese and Wiesenberger, 1982; Cummings *et al.*, 1989; Strieter *et al.*, 1999; Lysonski *et al.*, 1988). Another key theme focuses on the interdependencies between the traditional functions of the firm and their relationship to the external environment. These boundary spanning activities are considered a critical element linking information to the organization (Lysonski, 1985; Wood and Tandon, 1994; Lysonski and Woodside, 1989; Lysonski *et al.*, 1988). Lysonski (1985, p. 26) defines these boundary-spanning activities as "the interface between the firm and its market environment, as well as between departments in the firm". Internally these include the interdependencies between finance, engineering, production, marketing and sales, while externally include communication with customers, channel partners, specialist (e.g. advertising agencies) (Lysonski and Woodside, 1989), market analysts and competitors (Gorchels, 2005). Many authors believe that there has been an overemphasis on internal activities and that a reorientation towards external boundary spanning activities are needed to improve PM effectiveness (Tyagi and Sawhney, 2010; Murphy and Gorchels, 1996; Katsanis and Pitta, 1995). For a more detailed description of the PM within the firm, see Roach (2011).

Product management and SMEs

Even though the classic P&G model has been adopted by various SMEs (Katsanis *et al.*, 1996), this model is not considered well suited for smaller organizations who tend to be driven by daily concerns (Woodcock *et al.*, 2000). Thus, SMEs provide both a challenge and opportunity in the area of PM research. From one perspective, SMEs tend to lack new product development infrastructure, conduct insufficient competitive analysis

(Woodcock *et al.*, 2000) and commonly fail to translate solid research and development into successful innovations (O'Regan *et al.*, 2006). They tend to display strong technology-related PM practices (Huang *et al.*, 2002) but generally lack the ability to integrate many of the marketing-related PM tasks such as market research, market testing and customer intelligence. This integration tends to differentiate successful innovations, products and firms (Verhees and Meulenbergh, 2004; Huang *et al.*, 2002).

On the other hand, successful SMEs appear to have both formal and informal PM strategies (Karlsson and Olsson, 1998). Huang *et al.* (2002) suggest that the interrelationship of certain PM practices such as market research, analysis, product creation and commercialization were cited as the most crucial activities for product success. However, these PM activities were also described by the respondents as being the group of business activities that required the most improvement and attention. Gray *et al.* (1998) also found that one of the overarching PM capabilities – inter-functional coordination – had the strongest relationship with ROI suggesting that “sharing of market and product information ... and use of coordinated planning methods may be linked with improved business efficiency and profitability” (Gray *et al.*, 1998, p. 899). They go on to comment on the effectiveness of many of the PM activities including encouragement of customer input, after-sales emphasis, regular evaluation of value creation and systematic measurement of customer satisfaction.

The key difference between SMEs and their larger counterparts may lie in their simpler organizational structures (Becherer *et al.*, 2003) and ability to more quickly adapt to changing market needs (Hadjimanolis, 2000; Keskin, 2006). Their general flexibility and adaptability should facilitate greater boundary spanning activity between established functional areas. Since PM is essentially the general management of a product or product line and SMEs tend to have a limited range of products, they present a relatively constant and representative unit of measure to assess PM activity within the firm.

Product management, market orientation and performance

Recent research suggests that one of the most important factors in creating high-performance PM organizations was the reduction of organizational boundaries and the elimination of traditional functional silos (Tyagi and Sawhney, 2010). From this perspective, PM can be thought of as a holistic, interdependent, boundary spanning system (Katsanis and Pitta, 1995) which cuts through the traditional vertical specialties of the organization (Luck, 1969). This leads to looking at PM as a set of boundary spanning firm-level activities, which must be present in order to thrive in the marketplace.

What then is the relationship between MO and PM? MO entails the implementation of the marketing concept, by enhancing the firm's ability to anticipate, react and capitalize on environmental changes (Shoham *et al.*, 2005). Narver and Slater (1990) approach MO from a cultural perspective, while Jaworski and Kohli (1993) take a complimentary behavioural approach focusing on stakeholder relationships. Deshpandé *et al.* (1993) on the other hand initially defined MO as a “set of beliefs that puts the customer's interest first, while not excluding those of all other stakeholders such as owners, managers and employees” (Deshpandé *et al.*, 1993, p. 27). Later Deshpandé and Farley (1998) undertook an assessment of these three measurement scales finding that they were interchangeable in practice, reliable and valid, recommending a more parsimonious ten-item MORTN scale.

PM activities are however distinct from MO, since they focus on the firm's ability to extract external information and efficiently process it internally. For instance, a market-oriented firm may be culturally predisposed to customer centric activities, but may lack the capability to extract relevant customer needs and behaviour information from the marketplace. Even if the firm has this capability, it may lack the systems and processes to integrate this information into the organization for product-related decision making. Thus, market-oriented firms who lack PM abilities may be very efficient at external information gathering (for instance in SMEs the founder is often the primary information gatherer), but unable to adequately transfer this knowledge to the organization. The organization on the other hand, may be functionally very strong, but lack the ability to share and disseminate pertinent information across functional boundaries for use in decision making. Thus, PM goes beyond mere market-oriented culture and behaviour by enhancing (or mediating) its impact on firm performance. This study thus posits that boundary spanning PM behaviour may be one of the missing mediation effects between MO and performance as highlighted in several meta studies (Shoham *et al.*, 2005; Kirca *et al.*, 2005).

Clearly many authors believe that superior PM capabilities should positively affect firm performance. Yet, despite this general agreement, the relationship between firm performance and PM behaviour has hereto not been tested in any empirical way. Thus, for the purposes of model development, a 33-item survey instrument is adapted based on an investigation of the extant PM literature (Roach, 2011) as outlined in Appendix. For the purposes of model development, Deshpandé and Farley's (1998) ten-item MORTN scale was used as a measurement of the MO construct. These concepts are then integrated into a structural model, where PM mediates the relationship between MO and firm performance.

Methodology and model development

Sample and data collection

The model was tested on a sample of Canadian SMEs engaged in manufacturing. For the purpose of this study, SMEs were defined as:

- having greater than 5 and less than 250 employees; and
- being stand alone enterprises (i.e. not subsidiaries of larger entities).

The method of data collection was a self-reported on-line survey answered by senior managers of SMEs in May 2009. Questions were randomized using a seven-point Likert scale from "disagree completely" to "agree completely".

This method resulted in a 367 good responses of which 51 were deselected, since they self reported that they had less than 5 employees or greater than 250 employees. This left a useable sample of $n = 316$, which translates into a response rate of 3.81 per cent of the total population ($n = 8,295$). Of the respondents 219 (69.3 per cent) were chief officers, presidents or vice presidents of their companies. When senior managers were included this increased to 294 (93.1 per cent) of respondents. Of these respondents 228 (72.2 per cent) had greater than 15 years experience in their industry, while 183 (58.0 per cent) had greater than 20 years experience.

Construct development

The PM instrument was pre-tested on a sample of Atlantic Canadian SMEs ($n = 63$) (Roach, 2011). Minor modifications to the structure and wording of the

questions resulted, which included the addition of one new question (PM33) (see Appendix). The results suggested that the instrument was understandable, captured the PM concepts of interest and were interpreted correctly, thus establishing face validity.

On the basis of this pre-test and to further establish content validity, all of the PM and MO indicators were analyzed using principal component analysis (PCA). This confirmed the uni-dimensionality of the construct's indicators prior to further evaluation of reliability and validity (Götz *et al.*, 2010). Since PM is generally considered a marketing construct, both product management (33 – PM indicators) and market orientation (10 – MO indicators) were factor analyzed together. This approach was used to determine whether there was substantial correlation between both sets of measurement indicators.

Using a Varimax rotation, a visual analysis of the scree plot indicated a five-factor reduction as described in Table I. Product management indicators are denoted as “PM” while market orientation indicators are denoted as “MORes”.

This analysis revealed three “clean” product management (PM) factors (1, 4 and 5) and two market orientation factors (2 and 3). This indicates that market orientation and product management are measuring separate constructs, providing content validity. The market orientation MORTN scale generally maps to factors 2 and 3 with only two indicators (MORes2 and 3) not loading, while one product management indicator (PM12) loaded on the first market orientation factor. The latter is likely mimicking the market orientation indicator MORes6 and is thus considered redundant. However, given that most market orientation indicators loaded on the second and third factor, and to preserve this well-established parsimonious scale (Deshpandé and Farley, 1998), market orientation was initially maintained as a distinct single factor construct represented by all (and only) its well-established indicator variables.

In a similar fashion, three distinct product management factors (1, 4 and 5) were identified. These are interpreted as follows:

- PM factor 1 – channel analysis/support;
- PM factor 2 – product positioning; and
- PM factor 3 – marketing/technical integration.

These three factors are further developed in the following section. For a detailed list of the product management indicators, see Appendix.

Factor 1		Factor 2		Factor 3		Factor 4		Factor 5	
PM11	0.653	MORes5	0.723	MORes4	0.662	PM27	0.632	PM23	0.738
PM6	0.631	MORes9	0.686	MORes1	0.643	PM17	0.628	PM22	0.698
PM19	0.610	MORes6	0.673	MORes8	0.638	PM20	0.592	PM21	0.524
PM3	0.565	PM12	0.592	MORes7	0.599	PM26	0.520	–	–
PM5	0.483	MORes10	0.578	–	–	PM24	0.506	–	–
PM1	0.463	–	–	–	–	PM9	0.483	–	–
–	–	–	–	–	–	PM15	0.471	–	–

Table I.
Results of PCA after
Varimax rotation

Model specification and hypothesis development

The proposed PM model examines the relationship between MO, PM and firm performance (PERF). The generic PM model is presented below, with each latent construct represented as a reflective construct. In this model, MO is represented as an exogenous construct, while the three PM constructs are represented as mediating endogenous constructs (Figure 1).

The following hypotheses are proposed to test this model.

MO has a deep and rich history of research dating back more than 30 years, supporting the positive relationship between MO and firm performance (Deshpandé *et al.*, 1993; Jaworski and Kohli, 1993; Narver and Slater, 1990). Using the MORTN scale captures both the cultural and behavioural components of the MO construct in one parsimonious scale. The cultural component as espoused by Narver and Slater (1990), measures how well MO is embedded in the shared values of the organization, while the behavioural focus of Kohli *et al.* (1993) measures whether MO is rooted in the activities of the firm. These relationships have been widely supported through decades of research and multiple meta studies (Kirca *et al.*, 2005; Rodriguez *et al.*, 2004). For instance, Rodriguez *et al.* (2004) in their meta analysis found that the relationship between MO and business performance accounts for only about 12 per cent of the effect, but that it is positive and consistent world-wide. This relationship is also supported in the SME literature (Gray *et al.*, 1998, Pelham, 1997, Verhees and Meulenbergh, 2004). Thus, hypothesis one is:

H1. Market orientation (MO) positively affects firm performance.

Channel analysis and support (PM-F1) fundamentally measures external boundary spanning capabilities of the firm. It speaks to formal and informal systems which monitor the product in the marketplace, through both qualitative and quantitative feedback. This feedback is processed by the firm through analysis and used to make

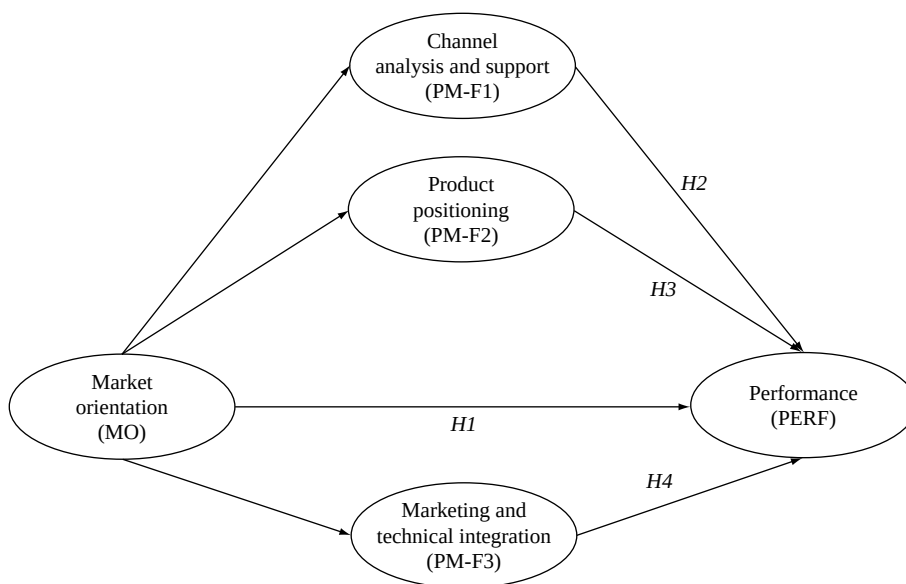


Figure 1.
Product management
model

decisions for such things as product phase out, pricing, sales support and forecasting. The efficiency of external boundary spanning activity should create the stage for more optimal decision making, which should ultimately be reflected in increased performance. Thus, hypothesis two is:

H2. Channel analysis and support (PM-F1) positively mediates the market orientation – firm performance relationship.

Product positioning (PM-F2) also measures external boundary spanning capability, which impacts the positional advantage of the product in the marketplace. This latent construct involves such activities as conducting research to benchmark products, obtaining customer feedback through primary research and conducting market tests. These activities are supported by appropriate business case analysis, which integrates such things as market forecasting and product feasibility. The external environment is also consistently monitored for product planning purposes. It measures the firm's ability to maintain and enhance its product position in the marketplace. Thus, hypothesis three is:

H3. Product positioning (PM-F2) positively mediates the market orientation – firm performance relationship.

Marketing and technical integration (PM-F3) on the other hand reflects a firm's internal boundary spanning capabilities as it relates to the interaction between the marketing and technical factions of the firm. It speaks to the permeability of internal functional interfaces; the sharing of ideas and information on an equitable basis. This communication is sustained by both formal and *ad hoc* systems, which support product initiatives such as concept development, refinement and development. When combined with a market-oriented culture it enhances the firm's ability to consistently deliver superior products. This cross-functional capability enhances the firm's product-market fit, which should lead to increased firm performance. Thus, hypothesis four is:

H4. Marketing and technical integration (PM-F3) positively mediates the market orientation – firm performance relationship.

Analysis and results

Method of analysis

Analysis was performed using Smart PLS (version 2.0.M3), a partial least squares (PLS) based structural equation modelling (SEM) program. This PLS-SEM program assesses the psychometric properties of the outer (or measurement) model, while simultaneously estimating the parameters of the inner (or structural) model (Al-Gahtani *et al.*, 2007). PLS appears to be gaining wider acceptability in management research including recent SME research (Brettel *et al.*, 2007), since it provides certain advantages over covariance-based SEM analysis. This includes the fact that PLS models are not constrained by the assumption of normality, are effective for smaller sample sizes and have the ability to handle both formative and reflective indicators. Unlike, covariance-based SEM, PLS models take into account measurement error through simultaneous regressions and thus do not employ goodness of fit measures as an indication of model robustness. It instead uses the significance of the path and the explained variance (R^2) of the model (Hulland, 1999). Since it is designed to explain variance, it is well suited for exploratory research, predictive applications and/or theory building (Genfen *et al.*, 2000).

Thus, the quality of any PLS model is based on its ability to satisfactorily describe the effects between latent constructs under investigation. As such, a two-tiered approach was applied which first examines the measurement model's quality, followed by an assessment of the extent to which the model reproduces the real data structure (i.e. the structural model) (Götz *et al.*, 2010).

The measurement model

The approach begins with the measurement of the outer model, which specifies the relationship between observed indicator variables and their underlying constructs. This process involves the assessment of indicator and construct reliability as well as convergent and discriminant validities.

Indicator reliability specifies which part of the indicator's variance can be explained by the related latent variable. About 50 per cent of an indicator's variance should be explained by the latent construct, indicating that the shared variance between an indicator and its construct is larger than the measurement error (Götz *et al.*, 2010). This manifests itself into an indicator loading of 0.70 or greater, however, weaker loadings are often observed when newly developed scales are used. Thus, for reflective indicators, loadings less than 0.40 should be eliminated from PLS measurement models (Götz *et al.*, 2010; Hulland, 1999). As a result, some PM indicators below this threshold were removed, which included PM11, PM16, PM20, PM27; resulting in all PM and performance indicators except for PM24 exceeding 0.70 (see italicised indicator loadings in Table II). Indicator loadings for MO on the other hand exceeded the 0.40 threshold, but several were below the rule of thumb of 0.70. Following precedence in recent research for the removal of low loading MO indicators to increase model goodness of fit (Narver *et al.*, 2004), low loading indicators were systematically removed until indicator's variance explained by the latent construct exceeded 0.50. The removed indicators included MORes1, 3, 4 and 8, all of which loaded below or near 0.60.

A common measure for examining convergent validity is the "average variance extracted" (AVE). It represents the indicator variance captured by the construct (including measurement error), relative to the total variance (Götz *et al.*, 2010). A threshold of greater than 0.5 indicates that there is more indicator variances than variance due to error. AVE also exceeded 0.50 for all constructs.

Indicator loadings and cross loadings for all of the latent constructs were then examined as per Table II. This established that all items load well on their respective constructs and, that even for indicators loading below 0.70, no cross loadings are indicated (Yoo and Alavi, 2001). In addition, each indicator loaded on their respective construct with high significance ($p > 0.0001$) based on the t-statistics of the outer model, which ranged from 10.52 to 51.78. These findings thus confirm the convergent validity of the latent constructs for the proposed PM model.

Measures of construct reliability reflect how well a construct is measured by their assigned indicators (Götz *et al.*, 2010). It is important that the indicators jointly measure the construct adequately, thus reliability results presented in Table III include both composite reliability and Cronbach's alpha. For the proposed PM model, all reliability measures easily exceeded the recommend thresholds of 0.70 (Bagozzi and Yi, 1988; Hair *et al.*, 2006; Nunnally and Bernstein, 1994), indicating strong correlation and establishing them as reliable measures of their respective latent constructs.

Table II.
Factor loadings
(italicised) and
cross loadings

	MO	PM-F1	PM-F2	PM-F3	PERF
MORes10	<i>0.724</i>	0.392	0.416	0.393	0.105
MORes2	<i>0.805</i>	0.543	0.467	0.459	0.206
MORes5	<i>0.845</i>	0.563	0.501	0.374	0.231
MORes6	<i>0.793</i>	0.580	0.466	0.319	0.151
MORes7	<i>0.587</i>	0.362	0.242	0.328	0.300
MORes9	<i>0.652</i>	0.362	0.417	0.332	0.180
PM1	0.475	<i>0.704</i>	0.537	0.388	0.219
PM19	0.487	<i>0.755</i>	0.536	0.488	0.293
PM3	0.435	<i>0.685</i>	0.426	0.384	0.215
PM5	0.477	<i>0.735</i>	0.436	0.461	0.344
PM6	0.470	<i>0.770</i>	0.458	0.440	0.249
PM20	0.470	0.550	<i>0.806</i>	0.397	0.263
PM24	0.369	0.404	<i>0.656</i>	0.364	0.134
PM9	0.499	0.598	<i>0.819</i>	0.434	0.248
PM17	0.368	0.381	<i>0.715</i>	0.289	0.184
PM21	0.485	0.596	0.504	<i>0.834</i>	0.302
PM22	0.359	0.421	0.292	<i>0.817</i>	0.290
PM23	0.365	0.423	0.411	<i>0.820</i>	0.231
PERFORM1	0.191	0.333	0.231	0.257	<i>0.836</i>
PERFORM2	0.209	0.333	0.255	0.328	<i>0.870</i>
PERFORM3	0.200	0.252	0.211	0.235	<i>0.715</i>
PERFORM4	0.146	0.217	0.133	0.208	<i>0.776</i>
PERFORM5	0.259	0.334	0.321	0.287	<i>0.807</i>
PERFORM6	0.226	0.248	0.165	0.278	<i>0.787</i>

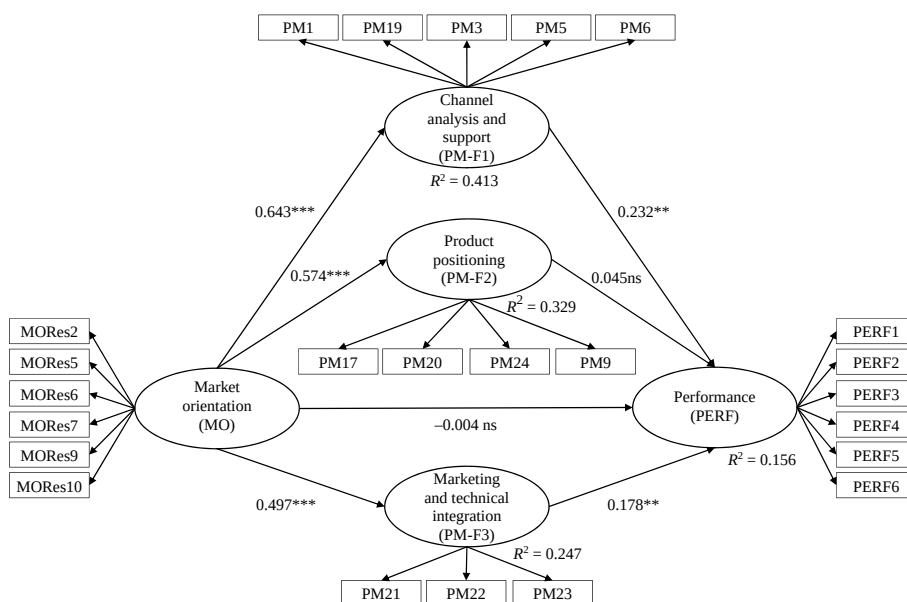
Table III.
Measurement model
assessment

Latent variable	AVE	CR	Cronbach's alpha	MO	PERF	PM-F1	PM-F2	PM-F3
MO	0.528	0.869	0.817	<i>0.727</i>	–	–	–	–
PERF	0.649	0.917	0.893	0.220	<i>0.806</i>	–	–	–
PM factor 1	0.528	0.848	0.776	0.616	0.256	<i>0.726</i>	–	–
PM factor 2	0.539	0.823	0.715	0.589	0.245	0.655	<i>0.734</i>	–
PM factor 3	0.680	0.864	0.767	0.549	0.244	0.577	0.488	<i>0.824</i>

Discriminant validity on the other hand refers to the dissimilarity between the measurement tool's ability to measure different constructs (Götz *et al.*, 2010). Thus, shared variance between the latent constructs' indicators must be larger than the variance shared with other latent variables (Hulland, 1999). The latent variable's AVE should be larger than the common variance (squared correlations) relative to any other of the model's constructs (Fornell and Larcker, 1981) in order to support discriminant validity. Table III indicates that the square root of the AVEs (e.g. elements italicised in the diagonals) are in all cases greater than the off-diagonal row and column elements. This supports the discriminant validity of the scales used.

Evaluation of the structural model

Figure 2 represents the generic PM model, and its associated indicator variables. The three product management factors, channel analysis/support (PM-F1),



Notes: ns = non-significant; * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$ using two-tailed t -tests

product positioning (PM-F2) and marketing/technical integration (PM-F3) are shown to mediate the relationship between MO and firm performance.

Beta loadings and their associated significance outlined in Figure 2. Since PLS models predict variance, R^2 values are included for each endogenous latent construct. All significant path coefficients are positive, indicating that their theoretical direction is correct, thus establishing nomological validity of the model.

Results

Support for the respective hypotheses was determined via the statistical significance of the path loadings. The strength of the model is based on the predicted R^2 of the endogenous constructs, specifically the predicted R^2 of firm performance.

The most significant finding in this study was that two of the three PM constructs fully mediated the MO – firm performance relationship. This result is somewhat unexpected given the substantial base of research on the positive relations between MO and performance (Gray *et al.*, 1998; Pelham, 1997; Verhees and Meulenbergh, 2004; Deshpandé *et al.*, 1993; Kirca *et al.*, 2005; Jaworski and Kohli, 1993; Narver and Slater, 1990; Rodriguez *et al.*, 2004). However, it does substantiate some of the SME literature, which questions MO's overriding importance in all conditions (Greenley, 1995; Pelham and Wilson, 1996; Becherer *et al.*, 2001; Frishammar, 2009). Thus, H_1 is rejected.

The two significant mediating constructs included channel analysis/support (PM-F1) and marketing/technical integration (PM-F3). The former measures the external boundary spanning activities of the firm, while the latter measures the internal coordination efforts between functional areas. Thus, H_2 and H_4 are supported. On the other hand, product positioning (PM-F2) had no effect on firm performance in the presence of PM-F1

Figure 2.
Product management model

and PM-F3. This insignificant relationship is also somewhat unexpected, since this construct is related to many of the key PM activities deemed to drive successful organizations and their performance. Such things as proper product benchmarking, strong market research, integration of customer feedback and business case analysis did not have a strong relationship with overall firm performance. Thus, H_3 is rejected.

Looking specifically at PM-F1 – channel analysis/support, this externally focused construct involves understanding market trends through obtaining, analyzing and acting upon channel information. This information processing is used by the firm to justify actions ranging from price adjustments to product line support. It involves proactively spanning the boundaries outside of the firm to gather relevant product information. Its impact on firm performance supports much of the PM literature, which emphasizes the importance of external boundary spanning activities (Lyonski, 1985; Tyagi and Sawhney, 2010; Lyonski and Woodside, 1989; Lyonski *et al.*, 1988). Conversely, PM-F3 – market/technical integration is an internal capability, which speaks to the cross-functional culture of the organization and its ability effectively coordinate activities between and within functional areas. This balanced approach between the marketing and technical factions of the organization is one of the cornerstones of the PM system. Interestingly, both the external information gathering (i.e. obtaining information as input to the decision process) and internal marketing/technical integration (i.e. the firm's ability to process information into decisions) together account for all of the explained variance (15.6 per cent) of firm performance.

Discussion

In this study, MO did not behave as predicted by the literature, having no effect on firm performance. This suggests that MO in the presence of certain aspects of PM behaviour, does not impact firm performance. These results are entirely due to the effect of two of the three PM constructs, namely the external boundary spanning capability of channel analysis/support (PM-F1) and the internal boundary spanning capability of marketing/technical integration (PM-F3). The other product management factor, product positioning (PM-F2), showed no power in this model. Thus, both of these external and internal PM capabilities appear to be good predictors of performance.

This outcome is academically important since this is the first study of its kind to investigate PM activities and their relationship to firm performance. The results support much of the PM literature, which infers that more emphasis on external boundary spanning should affect overall firm performance (Tyagi and Sawhney, 2010; Gorchels, 2005). Although some authors have measured various aspects of product manager's performance (Lyonski, 1985; Tyagi and Sawhney, 2010; Lyonski and Woodside, 1989; Lyonski *et al.*, 1988), this is the first study to link these external boundary-spanning activities directly to overall firm performance. In a similar fashion, there has been a significant amount of literature highlighting the benefits of cross-functional integration, specifically between the marketing and technological factions within the organization. Research on innovation has shown that improvements in SME's innovation practices can be achieved by beginning to structure organizations to allow for improved cross-functional and cross-disciplinary communication (Vermeulen, 2005). This evidence lends credibility to the importance of proper and effective PM activities within SMEs. It also provides empirical evidence that boundary spanning firm activities contribute positively to overall firm performance.

There may also be relevance for practitioners. Much has been written with respect to the development of a market-oriented culture within SMEs. Investments in building this culture and associated systems have been implemented by firms both large and small. However, this study indicates that the development of a market-oriented culture, may be restricted if firms do not invest in promoting external and internal boundary spanning activities. Firms with strong functional areas, content with merely analysing market information, may be limiting their performance by not encouraging external and/or internal boundary spanning activities. The ability of the firm to proactively garner feedback from the market about their products, conduct proper analysis and make informed decisions seems to separate higher performing SMEs in this study. When combined with strong internal integration between the marketing and technological factions, this appears to further increase firm performance. Good external information gathering, processed by an integrated multifunctional culture, results in the highest firm performance in this study.

Conclusions, implications and limitations

This study is exploratory in nature and thus care must be taken to not over-generalize these results. As a result, future research should investigate other firm capabilities related to both PM and MO. Specifically, the mediating effect of innovation on performance (Kirca *et al.*, 2005) and other aspects of MO, such as proactive MO (Narver *et al.*, 2004) and aspects of entrepreneurial orientation (Lumpkin and Dess, 1996), could add additional explanatory power to the model. Neither has any attempt been made to link the growing body of work on effectuation within the entrepreneurship literature (Frishammar, 2009; Sarasvathy and Dew, 2005; Sarasvathy, 2001; Morrish, 2009). Whether SMEs behave entrepreneurially through effectual networks to create new markets or merely conduct research to exploit predictable markets (the casual process) is yet to be explored. This research seems to generally indicate the latter in SMEs, but no controls for firm age or new ventures were used. How internal and external PM connections are rearranged to create artifacts such as new products, specifically in new ventures, would be an interesting avenue for future research.

In addition, objective, quantifiable measures of performance (e.g. ROI, ROA, etc.) could be added to support the subjective measures of firm performance used in this study. Multiple dimensions of performance could reduce the potential bias believed to result from the close relationship between some market-oriented behaviours and firm performance (Pelham, 1997). Lastly, the results of this study should be replicated in other geographic areas, for example with US or European SMEs. Using a more homogeneous population could also add additional insight to this area of research.

Notwithstanding the above concerns, this study has both academic and practical implications. First, it adds to the literature on the mediation effects between MO and firm performance, specifically in the area of SME orientation. According to Olson *et al.* (2005) how work is coordinated within the firm should receive more attention in the strategy implementation research. From a practical perspective, senior SME management may be able to improve firm performance by investing in the development of both external and internal PM capabilities. Although organizations large and small perform many of these PM boundary-spanning activities, those that excel at their integration should perform better and more easily adapt to their competitive environments.

Perhaps, the most salient finding in this study is that PM appears to be a good predictor of firm performance, even given the broad industry categories used in this Canada wide study. The full mediation between MO and firm performance adds to the empirical body of knowledge in the SME literature by suggesting that PM ability may account for a significant proportion of performance irrespective of how market orientated the firm may be.

PM is a broad construct whose tasks span the disparate functional areas of the firm. This research is the first study of its kind to link PM behaviour with firm performance. It establishes an empirical link between these boundary-spanning activities by relating the extant PM literature with that of MO. Perhaps, the time has come to rekindle academic interest in PM; an area of research that has languished in the management literature.

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Appendix

Product management measures

- PM01 We have strong systems for deciding which products and/or service opportunities to pursue or which to phase-out.
- PM02 Our product and/or service pricing (i.e. base price, discount schedules, etc.) are clearly communicated to all stakeholders.
- PM03 Our product and/or service pricing (i.e. base price, discount schedules, etc.) is determined via careful analysis and feedback from the marketplace.

- PM04 Our sales force or sales partners are consistently kept informed about our product and/or service strategy.
- PM05 Our sales force or sales partners consistently receive up to date product and/or service support (e.g. training, technical support, demonstrations and presentations).
- PM06 Sales data is consistently analyzed and measured against marketing programs and forecasts.
- PM07 We foster communication about our products and/or services within our firm in order to generate product improvement ideas.
- PM08 We foster communication about our products and/or services outside of our firm in order to generate product improvement ideas.
- PM09 Our product and/or service plans are supported by strong business case analysis.
- PM10 Our marketing budget is allocated by product and/or service category rather than generally at the firm level.
- PM11 We consistently monitor the profitability of each of our products and/or services.
- PM12 We consistently monitor our customer service by product and/or service category.
- PM13 Our firm consistently reviews customer comments, recalls and technical service data as input for future decisions about our products and/or services.
- PM14 Our firm consistently seeks customer input into the design of our products and/or services.
- PM15 Competitive research and analysis drives our product and/or service planning.
- PM16 We consistently scan our industry for new ideas, trends or phenomena to improve our product and/or service offerings.
- PM17 We systematically benchmark our product and/or service offerings against industry leaders.
- PM18 We have well-established systems for setting priorities for our product or service offerings.
- PM19 We have well-established systems for forecasting and monitoring product and/or service unit sales.
- PM20 We have well-established systems to monitor our market share by product and/or service category.
- PM21 We have well-established systems for involving business, marketing and technical personnel in our product and/or service efforts.
- PM22 Marketing and technical personnel communicate effectively and work well together when it comes to product and/or service issues.
- PM23 Marketing and technical personnel participate equally in developing new product and/or service concepts.
- PM24 We consistently use pre-test marketing (e.g. simulated market tests) to establish the viability of new products and/or services.
- PM25 We take a team approach to planning and developing our product and/or service promotional activities.

- PM26 We consistently monitor our external environment (e.g. social, political, economic, regulatory) in order to avoid surprises with respect to our products or services.
- PM27 We frequently conduct formal research studies to prove or disprove our pre-conceived notions about our products and/or services.
- PM28 We prefer to take a long-term approach to our product and/or service offerings rather than focusing merely on short-term objectives.
- PM29 Our products and/or services are mainly driven by technical superiority.
- PM30 We consistently monitor the effectiveness of our promotional activities by product and/or service category.
- PM31 Our products and services are mainly driven by market requests.
- PM32 External information is consistently communicated to internal staff for decision making about our products and/or services.
- PM33 Once a product or service is introduced to the market, we regularly perform an analysis of what went right and what went wrong, in an effort to continuously improve.

Market orientation measures

- MORes1 Our business objectives are driven primarily by customer satisfaction.
- MORes2 We constantly monitor our level of commitment and orientation to serving customer needs.
- MORes3 We freely communicate information about our successful and unsuccessful customer experiences across all business functions.
- MORes4 Our strategy for competitive advantage is based on our understanding of customers needs.
- MORes5 We measure customer satisfaction systematically and frequently.
- MORes6 We have routine or regular measures of customer service.
- MORes7 We are more customer-focused than our competitors.
- MORes8 I believe this business exists primarily to serve customers.
- MORes9 We poll end-users at least once a year to assess the quality of our products and services.
- MORes10 Data on customer satisfaction are disseminated at all levels of the organization on regular basis.

Performance measure

- PERFORM1 Over the past year, our sales growth was _____ when compared to our competitors.
- PERFORM2 Over the past year, our profit growth was _____ when compared to our competitors.
- PERFORM3 Over the past year, our employment growth was _____ when compared to our competitors.

- PERFORM4 Over the past year, the overall performance of the business met expectations.
- PERFORM5 Over the past year, the overall performance of the business exceeded that of our major competitors.
- PERFORM6 Over the past year, top management was very satisfied with the overall performance of the business.

About the author

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